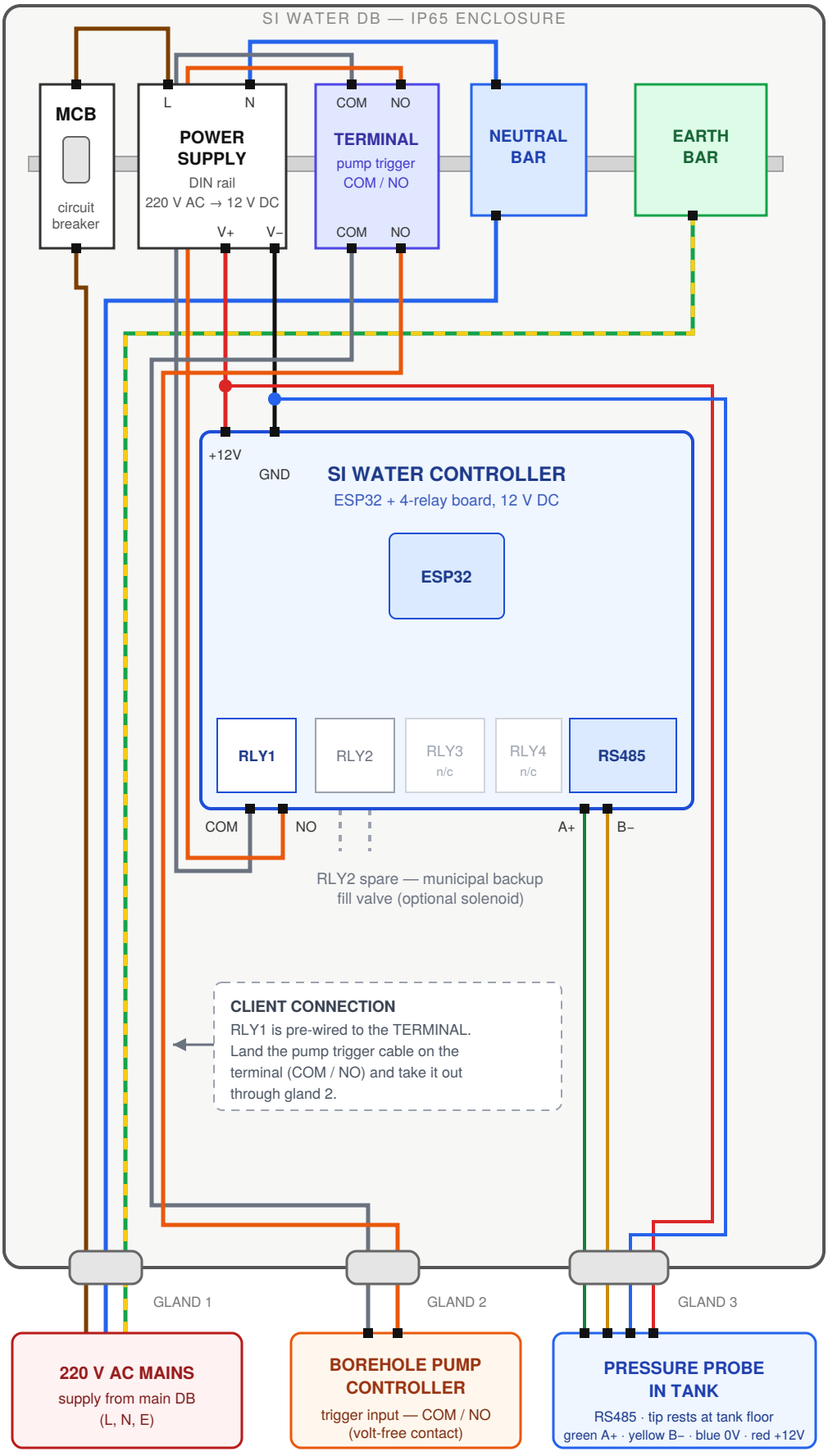
SI WATER — DB WIRING DIAGRAM

Solaire Intelligence SI Water · tank level monitoring & borehole pump control

WARNING: ISOLATE THE 220 V SUPPLY AT THE MAIN DB BEFORE OPENING THIS ENCLOSURE



WIRE LEGEND

- 220 V Live (brown)
- 220 V Neutral (blue)
- Earth (green/yellow)
- +12 V DC (red)
- 0 V DC (black)
- Relay COM — dry contact
- Relay NO — dry contact
- RS485 A+ (green)
- RS485 B− (yellow)
- Probe 0 V (blue)
- Dot = wires joined

Crossing wires are NOT joined.

NOTES

1. Isolate the 220 V supply at the main DB before opening or working in this enclosure.
2. 220 V AC feed enters via gland 1: Live to the MCB, Neutral to the neutral bar, Earth to the earth bar.
3. The DIN-rail PSU converts 220 V AC to 12 V DC and powers the controller board and the probe.
4. Relay 1 (COM / NO) is pre-wired to the DIN-rail TERMINAL. Land the pump trigger cable on this terminal, out via gland 2. Volt-free contact — switch the pump contactor / enable, never the motor directly.
5. Relay 2 is spare — reserved for the optional municipal backup fill valve (solenoid). RLY3–4 not used.
6. The pressure probe hangs in the TANK, tip at the tank floor. It is powered at 12 V DC straight from the PSU output (same V+ / V− terminals as the board) — it will not read without 12 V.
7. RS485: match A→A, B→B per the tag on the probe cable. If readings stay blank, swap A and B. Twisted pair; keep clear of mains cabling.
8. Leave the probe's vented cable end inside the dry enclosure — do not seal or submerge it.
9. All logic runs locally. Dashboard: si-water-XXXXXX-01.local (your unit's Client ID) · portal: portal.solaire-intelligence.co.za

SI WATER DB — WIRING DIAGRAM Doc DB-001 · Generic reference — all SI Water units Solaire Intelligence · solaire-intelligence.co.za · sales@solairesa.co.za	DRAWN Solaire Intelligence July 2026	REVISION **1.0**	SHEET **1 of 1**